When High Accuracy Isn't Enough: An Artifact-Aware Comparison of Classical and Embedding-Style Approaches to Misinformation Detection

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GitHub repository: https://github.com/jessy0017/dcit316-misinformation-detection-ghana

Abstract

This paper investigates misinformation detection using the ISOT Fake and Real News Dataset, comparing a classical TF-IDF and Logistic Regression baseline against an embedding-style TF-IDF+SVD approach, with particular attention to dataset-specific artifacts that may inflate reported performance. Exploratory analysis revealed two significant sources of label leakage: a perfect leak via the article subject metadata field, and a near-total leak via the literal string "(Reuters)" present in 99.2% of real articles but only 0.04% of fake articles, consistent with the dataset's documented source-specific collection methodology. The classical baseline achieved 98.68% accuracy on unmodified text, dropping modestly to 98.00% after the identified artifact was removed, while the embedding-style approach achieved 96.55% accuracy on the same artifact-stripped text. A quantified error analysis of 156 disagreement cases found the classical baseline correct in 91% of them, with 56% of disagreement cases featuring capitalized, sensationalist headline markers, suggesting stylistic cues meaningfully influence classifiability in this dataset. These findings demonstrate the importance of artifact-aware evaluation in misinformation detection research and caution against interpreting high benchmark accuracy as evidence of genuine, generalizable detection capability, particularly when considering adaptation to lower-resource, Ghanaian or African social media contexts.

Introduction

Misinformation on digital and social platforms has become one of the most pressing challenges facing information ecosystems worldwide. As internet and smartphone penetration continues to

grow across Ghana and the wider African region, social media platforms are increasingly serving as primary channels through which citizens consume news and form opinions on politics, public health, and civic issues. This growing reliance makes the ability to detect and understand misinformation-style content a matter of practical importance, not merely an academic exercise.

While large-scale, platform-specific datasets for African or Ghanaian misinformation remain scarce due to current API restrictions and data access constraints, the methodological questions underlying misinformation detection, how well do simple, explainable models perform compared to more complex representations, and what dataset-specific shortcuts might inflate apparent performance, are directly transferable and worth investigating using available, well-documented resources before attempting deployment on lower-resource, harder-to-access Ghanaian social text.

This paper investigates two complementary research questions. First, how does a classical, interpretable baseline (TF-IDF combined with Logistic Regression) compare against an embedding-style representation (TF-IDF combined with Truncated Singular Value Decomposition) for binary misinformation-style text classification? Second, and more critically, to what extent does apparent model performance on a benchmark misinformation dataset reflect genuine content-based signal versus superficial, dataset-specific artifacts that would not generalize to real-world settings?

To address these questions, this study uses the publicly available ISOT Fake and Real News Dataset, a widely cited benchmark in the fake news detection literature. Through exploratory data analysis, this work identifies two significant sources of potential label leakage in the dataset: a near-total leak via the subject metadata field, and a near-total leak via the literal string "(Reuters)" appearing in the article body of real news articles. Rather than treating these findings as incidental preprocessing notes, this paper treats them as a central part of the investigation, deliberately constructing an artifact-stripped evaluation to assess how much of the classifiers' reported performance survives once these shortcuts are removed.

The key findings of this work are threefold. First, a classical TF-IDF and Logistic Regression baseline achieves 98.68% accuracy on the unmodified dataset, which drops only modestly to 98.00% once the identified "(Reuters)" artifact is removed, suggesting the classifier retains

meaningful signal beyond this single superficial marker. Second, the embedding-style TF-IDF+SVD model underperforms the classical baseline (96.55% accuracy), indicating that dimensionality reduction into a small number of dense components discards discriminative lexical signal that TF-IDF's high-dimensional sparse representation preserves. Third, a quantified error analysis of disagreement cases between the two models reveals that the classical baseline is substantially more accurate on cases involving sensationalist, capitalization-heavy headline style, suggesting that stylistic cues not simply topical content play a measurable role in distinguishing the two classes within this dataset.

The remainder of this paper is structured as follows. Section 2 reviews related work in fake news detection and classical versus embedding-based text classification. Section 3 describes the dataset used, including its provenance and known limitations. Section 4 details the preprocessing pipeline and modeling methodology. Section 5 presents quantitative results. Section 6 discusses the implications of the identified dataset artifacts and the observed model behavior. Section 7 outlines the limitations of this study, particularly regarding generalizability to Ghanaian and African social media contexts. Section 8 presents an ethics statement, and Section 9 concludes with a summary of contributions and directions for future work.

Related Work

Automated fake news and misinformation detection has been an active area of research for close to a decade, drawing on natural language processing, network analysis, and increasingly, large language models. Shu, Sliva, Wang, Tang, and Liu (2017) provide one of the foundational surveys framing fake news detection as a data mining problem spanning content-based features, social context, and propagation patterns. Around the same period, several benchmark datasets emerged to support supervised learning approaches, including the LIAR dataset of short political statements (Wang, 2017) and the ISOT Fake and Real News Dataset (Ahmed, Traore, & Saad, 2017), the latter of which is used in this study. Zhou and Zafarani (2020) later synthesized this growing body of work into a broader survey, distinguishing between content-based, propagationbased, and source-based detection paradigms.

Within content-based approaches, a persistent methodological question is how classical, interpretable representations compare against denser, semantically-oriented representations. Classical approaches typically rely on bag-of-words or TF-IDF features paired with traditional

classifiers such as Logistic Regression, Naive Bayes, or Support Vector Machines (Pérez-Rosas, Kleinberg, Lefevre, & Mihalcea, 2018). Dimensionality-reduction techniques such as Truncated Singular Value Decomposition or non-negative matrix factorization offer an intermediate step between sparse lexical representations and full neural embeddings, compressing highdimensional text features into a smaller set of latent components intended to capture underlying semantic structure. Lee and Seung (1999), in foundational work on non-negative matrix factorization, demonstrate that such decomposition techniques can recover interpretable, partsbased latent structure from high-dimensional data, though the fidelity of this compression depends heavily on the underlying data's structure, a consideration directly relevant to this study's finding that only 20% of TF-IDF variance was retained by 100 SVD components.

A smaller but methodologically important thread of literature focuses not on maximizing detection accuracy, but on interrogating whether reported performance on benchmark datasets reflects genuine content understanding or superficial dataset artifacts. This concern echoes a broader challenge in computational social science: Vespignani (2009) argues that predicting the behavior of techno-social systems is fundamentally constrained by the difficulty of disentangling genuine social signal from artifacts of data collection and measurement, particularly as automated systems are deployed at scale without adequate scrutiny of the data they are trained on. Applied to fake news detection specifically, this concern manifests as a risk that classifiers trained on curated benchmark datasets, which often draw fake and real articles from different sources with distinct formatting conventions, may learn to exploit these formatting differences (such as source-specific datelines or metadata) rather than developing genuine sensitivity to misleading or fabricated content.

A related epistemic caution comes from outside the fake-news literature specifically: Dhar (2013) argues that predictive accuracy, not explanatory elegance, should be the primary criterion by which data-driven models are judged, but cautions that machine learning methods "pick up noise" in data with "no way to distinguish between signal and noise" unless researchers actively interrogate what a model has actually learned. Dhar further observes that simpler models tend to be more robust to future, out-of-sample data than more complex ones. Both observations are directly relevant to this study: the modest accuracy drop after removing an obvious dataset artifact (Section 6.1), and the finding that the simpler TF-IDF baseline outperformed the

compressed embedding-style model (Section 6.2), are concrete illustrations of exactly the caution Dhar raises at a general, cross-disciplinary level.

This study positions itself within this second thread. Rather than reporting a single headline accuracy figure, it deliberately investigates the ISOT dataset for exactly this kind of artifact, quantifies its impact on reported performance, and compares a classical TF-IDF baseline against an SVD-based embedding-style approach under both artifact-present and artifact-stripped conditions. This combination of methodological comparison and artifact-aware evaluation distinguishes this work from studies that report benchmark accuracy without interrogating its sources, and contributes a concrete, reproducible case study relevant to researchers considering the ISOT dataset, or similarly constructed benchmark datasets, for future work, including eventual adaptation to Ghanaian and African social media contexts where analogous sourcebased artifacts may exist in different forms.

Data

This study uses the ISOT Fake and Real News Dataset (Ahmed, Traore, & Saad, 2017), obtained via a public Kaggle mirror of the original University of Victoria ISOT Research Lab release. The dataset comprises 44,898 English-language news articles collected primarily between 2016 and 2017, with 23,481 labeled as fake and 21,417 labeled as real (a roughly balanced 52.3%/47.7% split). Each record includes an article title, full body text, a subject category, and a publication date.

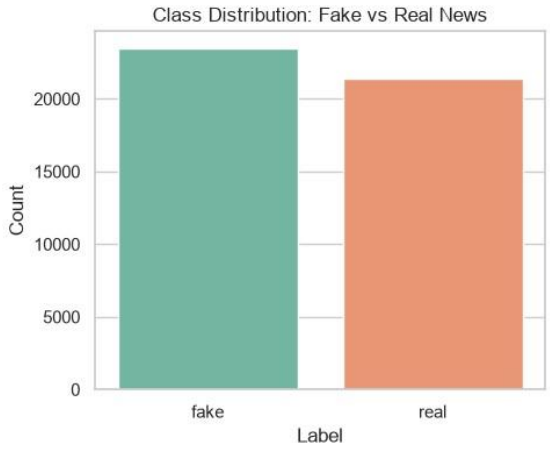
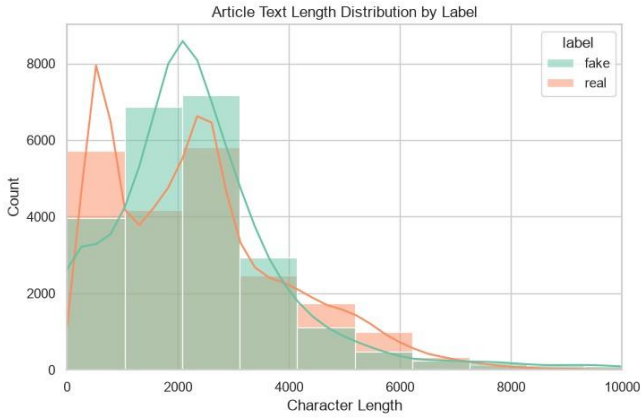


Figure 1: Class distribution fake (23,481) vs. real (21,417) articles.

According to the dataset's documented construction methodology, real articles were collected specifically by crawling Reuters.com, while fake articles were sourced from a range of websites flagged as unreliable by fact-checking organizations such as PolitiFact. This asymmetric collection process real and fake articles originating from structurally different sources rather than being drawn from a shared population and subsequently labeled has direct implications for this study, discussed further in Sections 5 and 6.

Preprocessing

Prior to modeling, 670 records (1.5% of the dataset) with article bodies shorter than 20 characters were removed as likely empty or corrupted entries, leaving 44,228 usable records. Article title and body text were concatenated into a single text field for feature extraction. The subject and date fields were deliberately excluded from modeling features following exploratory analysis, which revealed that subject perfectly predicts the label and would therefore constitute an unrealistic shortcut rather than genuine content-based signal.



label	count	mean	std	min	25%	50%	75%	max
fake	23481.0	2547.396235	2532.884399	1.0	1433.0	2166.0	3032.0	51794.0
real	21417.0	2383.278517	1684.835730	1.0	914.0	2222.0	3237.0	29781.0

Figure 2: Article text length distribution by label, showing greater variability among fake articles.

Exploratory Findings: Two Label Leaks

Two significant sources of unintended label leakage were identified during exploratory data analysis.

Subject metadata leak. Cross-tabulating the subject field against label revealed a perfect partition: articles labeled politicsNews or worldnews were exclusively real (21,417 articles, matching the real class total precisely), while articles labeled News, politics, left-news, Government News, US_News, or Middle-east were exclusively fake. This is consistent with the documented collection process, in which real and fake articles were scraped using different source-specific category taxonomies rather than drawn from a unified labeling process.

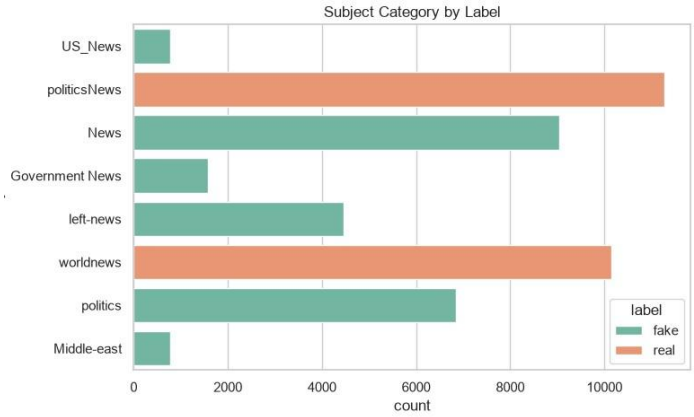


Figure 3: Subject category by label, showing a perfect partition between real (politicsNews, worldnews) and fake (all other categories) subjects.

Source-attribution string leak. Further inspection revealed that 99.2% of real articles contain the literal substring "(Reuters)" within the article body, a standard wire-service dateline convention (e.g., "WASHINGTON (Reuters) -"), compared to only 0.04% of fake articles. This is a direct artifact of the asymmetric collection process described above: because real articles were crawled specifically from Reuters.com, the outlet's own formatting convention becomes, in effect, a proxy label.

Both findings indicate that a classifier trained naively on this dataset risks learning sourceformatting artifacts rather than features genuinely indicative of misinformation. This motivates the artifact-aware evaluation methodology described in Section 4.

Known Limitations of the Dataset

As documented in the accompanying data card (data_card.md), this dataset consists of full news articles rather than short-form social media posts, and its content is US-centric in both topic and sourcing. Findings from this dataset should therefore not be assumed to generalize directly to

Ghanaian or African social media misinformation without further validation on locally representative data, a limitation discussed further in Section 7.

Methods

Feature Representation

Following the preprocessing steps described in Section 3.1, the combined title-and-text field for each article was used as the sole textual input to two feature representation approaches.

TF-IDF representation. Term Frequency-Inverse Document Frequency (TF-IDF) vectorization was applied using scikit-learn's TF-IDFVectorizer, restricted to the 10,000 most frequent terms with English stop words removed. This produces a high-dimensional, sparse representation in which each document is characterized by the relative importance of individual vocabulary terms.

Embedding-style representation. To approximate a lower-dimensional, semantically-oriented representation without requiring transformer infrastructure unavailable in the coursework environment, Truncated Singular Value Decomposition (TruncatedSVD) was applied to the TF-IDF matrix, reducing it to 100 dense components. This mirrors the classical-baseline-versusembedding-style comparison methodology used in prior coursework (DCIT 316 Individual Assignment 5) and is consistent with established practice in using SVD-based latent semantic representations as a computationally lightweight proxy for dense embeddings (Lee & Seung, 1999).

Models

Both feature representations were paired with a Logistic Regression classifier (scikit-learn, max_iter=1000), trained separately on each representation:

i. Classical Baseline: TF-IDF features -> Logistic Regression

ii. Embedding-style Comparison: TF-IDF -> TruncatedSVD (100 components) -> Logistic Regression

All models used random_state=42 for reproducibility.

Artifact-Aware Evaluation Design

Given the source-attribution leak identified in Section 3.2, this study evaluates the classical baseline under two conditions to isolate the effect of the identified artifact:

i. With artifact: trained and evaluated on the original text, including any occurrence of "(Reuters)" and its surrounding dateline pattern.

ii. Artifact-stripped: trained and evaluated on text with the "(Reuters)" marker and its typical preceding dateline pattern (e.g., "CITY (Reuters) -") programmatically removed via regular expression substitution prior to vectorization.

The embedding-style model was trained and evaluated exclusively on the artifact-stripped text, ensuring a fair, like-for-like comparison against the artifact-stripped classical baseline rather than against the inflated with-artifact figure.

Train/Test Split

Data was split into 80% training and 20% test sets using stratified sampling on the label field (train_test_split, random_state=42, stratify=y), ensuring both sets preserved the original class distribution. The same random seed and split logic were applied consistently across the artifactpresent, artifact-stripped, and embedding-style evaluation conditions to enable direct comparison.

Evaluation Metrics

Model performance was assessed using accuracy, macro-averaged F1 score, and per-class precision/recall/F1, computed via scikit-learn's classification_report. Macro F1 was prioritized as the primary comparison metric alongside accuracy, since it weights both classes equally regardless of class imbalance, which is relevant given the dataset's approximately 52%/48% split.

Error and Disagreement Analysis

To move beyond aggregate metrics, a case-level disagreement analysis was conducted between the artifact-stripped classical baseline and the embedding-style model. Test set predictions from

both models were compared directly, and instances where the two models disagreed were isolated for manual and quantitative inspection. For each disagreement case, the true label and each model's prediction were recorded, allowing identification of which model was correct more often, and manual review of a sample of disagreement texts was used to identify qualitative patterns, which were subsequently quantified using a regular-expression check for capitalized, sensationalist headline markers.

Results

Dataset Overview

After preprocessing, the dataset comprised 44,228 articles. Class balance was approximately 51.6% fake and 48.4% real in both the training and test splits, consistent with stratified sampling. Figure 1 shows the overall class balance. Figure 2 shows that fake articles exhibited greater variability in length (SD ≈ 2,533 characters, max 51,794) compared to real articles (SD ≈ 1,685 characters, max 29,781), though mean lengths were broadly similar (2,547 vs. 2,383 characters).

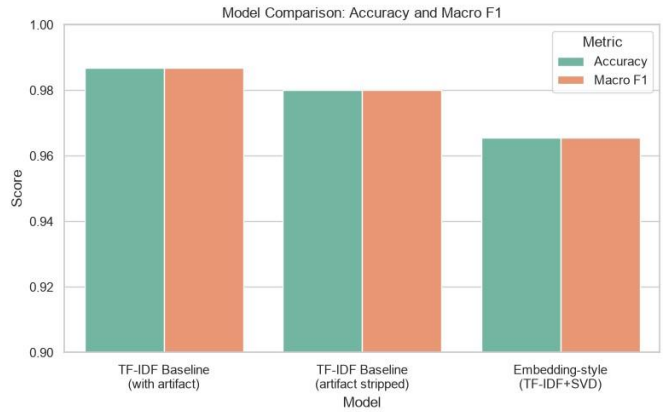
Subject and Source-Attribution Leak

As detailed in Section 3.2, the subject field was found to perfectly partition the label (Figure 3), and 99.2% of real articles contained the "(Reuters)" substring compared to 0.04% of fake articles.

Model Performance

Model	Accuracy	Macro F1
TF-IDF Baseline (with Reuters artifact)	0.9868	0.9868
TF-IDF Baseline (artifact-stripped)	0.9800	0.9800
Embedding-style (TF-IDF + SVD, 100 comps)	0.9655	0.9655

Table 1: Model comparison across evaluation conditions.



	Model	Accuracy	Macro F1
0	TF-IDF Baseline\n(with artifact)	0.986774	0.986765
1	TF-IDF Baseline\n(artifact stripped)	0.979991	0.979978
2	Embedding-style\n(TF-IDF+SVD)	0.965521	0.965503

Figure 4: Accuracy and Macro F1 comparison across all three evaluation conditions.

Removing the "(Reuters)" artifact reduced the classical baseline's accuracy by only 0.68 percentage points (98.68% to 98.00%), a smaller drop than might be expected given how strongly the artifact correlates with the label in isolation. This suggests the classifier retains substantial discriminative signal from the remaining text even after the most obvious shortcut is removed, though it does not rule out subtler, unexamined artifacts (Section 6.1).

The embedding-style model, trained and evaluated exclusively on artifact-stripped text for fair comparison, underperformed the artifact-stripped classical baseline by 1.45 percentage points (98.00% vs. 96.55%). The 100-component SVD representation retained only 20.16% of the original TF-IDF matrix's explained variance, indicating substantial information loss during dimensionality reduction.

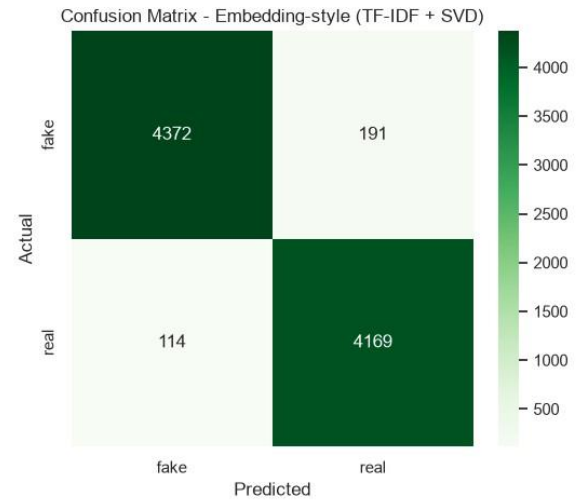


Figure 5: Confusion matrix for the embedding-style (TF-IDF+SVD) model.

Confusion matrices for all three conditions show that misclassification was distributed relatively evenly between false positives and false negatives in each condition, with no strong bias toward either class.

Error and Disagreement Analysis

Comparing predictions from the artifact-stripped classical baseline and the embedding-style model on the shared test set (n = 8,846) identified 156 disagreement cases (1.76% of the test set). Of these, the classical baseline was correct in 142 cases (91.0%), while the embedding-style model was correct in only 14 cases (9.0%).

Manual inspection of a sample of disagreement cases revealed a qualitative pattern: many articles the embedding-style model misclassified (while the baseline classified correctly) featured capitalized, sensationalist headline phrasing (e.g., "DISGUSTING!", "BREAKING:", all-capitals words such as "STUPIDLY" or "ILLEGAL"). This observation was quantified by checking for four-or-more-letter all-capitals words within the first 200 characters of each disagreement case's text: 88 of 156 disagreement cases (56.4%) contained such markers. This suggests the sparse, high-dimensional TF-IDF representation retains sensitivity to specific stylistic and lexical markers of sensationalism that the compressed 100-dimensional SVD representation does not preserve as effectively.

Discussion

Interpreting the Artifact-Stripping Result

The modest 0.68-point accuracy drop after removing the "(Reuters)" marker is notable. Given that this single substring alone would trivially predict the label with near-perfect accuracy for the 99.2% of real articles containing it, a much larger performance drop might have been expected if the classifier depended heavily on this feature. Two non-exclusive explanations are plausible. First, TF-IDF with 10,000 features and English stop-word removal likely down-weights a single short substring like "(Reuters)" relative to the broader vocabulary distinguishing the two classes, meaning its removal has a proportionally small effect on the overall feature space. Second, and more likely given the dataset's documented construction (Section 3), other systematic differences between Reuters' house style and the varied styles of the flagged fake-news sources, for instance,

sentence structure, punctuation conventions, or topic-specific vocabulary, may independently distinguish the classes, meaning the "(Reuters)" marker is only one of potentially several sourcecorrelated artifacts rather than the sole driver of classifier performance. As Dhar (2013) cautions, a model that "works" in the sense of achieving high accuracy does not by itself indicate what the model has actually learned; machine learning methods readily pick up noise or incidental correlations alongside genuine signal, with no built-in way to distinguish between them. This finding underscores that identifying and removing a single obvious artifact does not guarantee an artifact-free evaluation; genuinely robust misinformation detection would require training on genuinely comparable content sourced through a shared collection methodology, which the ISOT dataset, by design, does not provide.

Why the Classical Baseline Outperformed the Embedding-Style Model

The classical TF-IDF baseline outperforming the SVD-based embedding-style model is consistent with the low explained variance (20.16%) retained by the 100-component SVD reduction. This indicates that a substantial portion of the discriminative information present in the original 10,000-dimension TF-IDF space was discarded during compression. The disagreement analysis (Section 5.4) supports this interpretation: sensationalist stylistic markers, which are naturally represented as distinct, high-weight terms in a sparse TF-IDF vector, appear to be diluted or lost when compressed into a small number of dense latent components optimized to capture overall variance rather than class-discriminative signal specifically. This result is also consistent with Dhar's (2013) observation that simpler, sparser models tend to be more robust than more complex ones, here, the higher-dimensional but simpler TF-IDF representation outperformed the more heavily transformed embedding-style representation. This result should not be read as a general claim that embedding-based methods underperform classical methods in misinformation detection broadly; rather, it reflects a specific limitation of using TF-IDF+SVD as a lightweight embedding proxy, as opposed to a genuine contextual embedding model such as a transformer, which was outside the scope of the computational resources available for this coursework project.

Implications for Real-World and Ghana-Context Application

The scale of the identified label leakage in the ISOT dataset carries an important implication for anyone considering this dataset, or similarly constructed benchmark datasets, for real-worldoriented misinformation research: high reported accuracy on such benchmarks should not be interpreted as evidence that a model has learned to detect misinformation in a generalizable sense. A model trained on this dataset would likely perform far worse if deployed on genuinely comparable real and fake content drawn from the same publication source and formatting conventions, which is precisely the harder, more realistic scenario a Ghanaian or African social media misinformation detection system would need to handle, since such systems cannot rely on a convenient source-identity shortcut to distinguish misleading from legitimate posts circulating within the same platform.

Limitations

- Dataset domain mismatch. This study is grounded entirely in a US-centric, Englishlanguage news article dataset. No Ghanaian or African social media text was used, and findings should not be assumed to generalize to code-mixed, shorter-form, or dialectinflected social media content typical of Ghanaian platforms without further validation.

- Embedding proxy limitation. The "embedding-style" model in this study uses TF-IDF combined with Truncated SVD rather than a genuine contextual embedding model (e.g., a transformer-based sentence embedding). This was a deliberate scope decision given available computational resources, but it means the reported underperformance of the embedding-style approach should not be generalized to modern transformer-based embeddings, which typically capture semantic relationships more effectively than linear dimensionality reduction of sparse lexical features.

- Residual, unexamined artifacts. While this study identified and addressed two significant label leaks (subject and the "(Reuters)" substring), the modest accuracy drop after artifact-stripping suggests other, subtler source-correlated artifacts may remain unaddressed (Section 6.1). The reported artifact-stripped accuracy (98.00%) should

therefore be treated as an upper bound rather than a realistic estimate of real-world misinformation-detection performance.

- Single dataset, single split. All results are based on a single train/test split of a single dataset. No cross-validation or evaluation on an independent, out-of-domain test set was performed, which limits confidence in the stability of the reported metrics.

- Predictive, not causal. Consistent with Dhar's (2013) distinction between predictive and causal modeling, this study makes no causal claims about why articles are fake or real; it evaluates predictive classification performance only. No human annotation or manual fact-checking was performed as part of this study; all labels are inherited directly from the original dataset compilers.

Ethics Statement

This study uses only a publicly available, pre-labeled benchmark dataset (ISOT Fake and Real News Dataset) and does not involve the collection, scraping, or processing of any personal, private, or identifiable social media user data, in compliance with course requirements. No new data was collected from social media platforms.

The models developed in this study are intended strictly for educational and research purposes and are explicitly not intended for deployment as an automated content moderation or misinformation-flagging system. Given the identified dataset artifacts and the resulting risk that reported accuracy overstates real-world generalizability (Sections 6 and 7), deploying a model trained on this dataset to make unilateral classification decisions about real-world content would risk both false positives (incorrectly flagging legitimate content) and false negatives (failing to catch genuine misinformation), either of which could cause real harm if used without human oversight, particularly in a Ghanaian/African context for which this dataset was not designed or validated.

AI assistance was used in this project for code debugging, explanation, and drafting support; a full disclosure of AI tool usage, prompts, and verification steps is provided in the accompanying ai_use_disclosure.md file. All code, results, and interpretations presented in this paper were reviewed, run, and verified by the author.

Conclusion

This study set out to compare a classical TF-IDF-based baseline against an embedding-style TF-IDF+SVD approach for misinformation-style text classification, while critically examining whether reported performance on a widely used benchmark dataset reflects genuine contentbased detection capability. The investigation found that the classical baseline (98.00% accuracy after artifact removal) outperformed the embedding-style approach (96.55% accuracy), and that a substantial portion of the dataset's apparent classifiability stems from source-attribution artifacts, most notably a near-total label leak via the literal "(Reuters)" substring present in 99.2% of real articles. Quantified error analysis further revealed that the classical baseline's advantage was concentrated in cases involving sensationalist, capitalization-heavy headline style, suggesting stylistic rather than purely topical signal plays a measurable role in this dataset's classifiability.

These findings contribute a concrete, reproducible case study demonstrating the importance of artifact-aware evaluation in misinformation detection research, and caution against treating high benchmark accuracy on datasets like ISOT as evidence of genuine, generalizable misinformation-detection capability. Future work should extend this artifact-aware methodology to genuinely Ghanaian and African social media text, ideally sourced such that real and misleading content share a common collection methodology to avoid the kind of sourcecorrelated shortcuts identified here. Future work should also replace the TF-IDF+SVD embedding proxy used in this study with a genuine transformer-based contextual embedding model, to assess whether the observed underperformance of the embedding-style approach is specific to this lightweight proxy or reflects a more general finding.

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